



Theory Question Paper

National Physics Olympiad 2024

Category D

Time: 2 Hours

23 February, 2024

Name :
Registration No :
Class (In 2023) :
Institute :
Phone No :
Email :

General Instructions

READ THESE INSTRUCTIONS FIRST

- Make sure that Participant Name, Registration number, Class, Name of Institution and Contact number are written clearly.
- Answer all the questions. There are no optional questions. You need to attempt all the questions.
- Question parts are often independent of one another. Please read the entire question first.
- If you assume any unknown answer of the previous questions and correctly carry out the calculations of later questions, you will be awarded partial marks.
- There is no additional penalty for incorrect answers. Incomplete solutions with a proper approach will carry partial marks.
- All the data is given in English. In case of any confusion or inaccuracies regarding the questions and also if you are having trouble understanding the language of the questions, ask the invigilator.
- Only for the D category, there is no Bangla version in the question paper.
- Scientific calculator is allowed.
- All rough work must be demonstrated and recorded on the pages specified for rough work and on the blank pages attached at the end of this question paper.
- Cell phones and any other electronic devices are strictly prohibited for entire the exam duration. You may not use any tables, books, or collections of formulas except this paper.
- All the necessary values of the constant are given in the **Constant Table**.
- This question paper may not be used (for commercial purposes) after the competition without the permission of the Bangladesh Physics Olympiad.

Natural Constants

Acceleration due to gravity	g	$= 9.81 m \cdot s^{-2}$
Boltzmann constant	k_B	$= 1.38 \times 10^{-23} J \cdot K^{-1}$
Current Mass of the Sun	M_s	$= 2.00 \times 10^{30} kg$
Current Radius of the Sun	R_s	$= 7.00 \times 10^8 m$
Magnitude of the electron charge	e	$= 1.60 \times 10^{-19} C$
Mass of the electron	m_e	$= 9.11 \times 10^{-31} kg$
Mass of the proton	m_p	$= 1.67 \times 10^{-27} kg$
Mass of the neutron	m_n	$= 1.67 \times 10^{-27} kg$
Permeability of free space	μ_0	$= 1.26 \times 10^{-6} T \cdot m \cdot A^{-1}$
Permittivity of free space	ϵ_0	$= 8.85 \times 10^{-12} F \cdot m^{-1}$
Planck's constant	h	$= 6.63 \times 10^{-34} J \cdot s$
Avogadro Constant	N_A	$= 6.02 \times 10^{23} mol^{-1}$
Speed of light in vacuum	c	$= 3 \times 10^8 m \cdot s^{-1}$
Stefan-Boltzmann constant	σ	$= 5.67 \times 10^{-8} W \cdot m^{-2} \cdot K^{-4}$
Universal gas constant	R	$= 8.31 J \cdot mol^{-1} \cdot K^{-1}$
Universal Gravitational constant	G	$= 6.67 \times 10^{-11} N \cdot m^2 \cdot kg^{-2}$
Wien's constant	b	$= 2.90 \times 10^{-3} m \cdot K$
ln 2	$\approx$	$= 0.69$
ln 3	$\approx$	$= 1.10$
ln 10	$\approx$	$= 2.30$
Base of the Napierian logarithm e	$\approx$	$= 2.72$

Problem 1: A 2-body System! [13 Points]

We consider a two-body system comprising two point masses, labelled as m_1 and m_2 , positioned at $\vec{x}_1$ and $\vec{x}_2$ respectively. They exclusively interact with each other; the only force acting on each object is the gravitational force due to the other.

A	Write the Newton's law for each individual body using the above variables.	1 pts
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From these equations show that:

B	$\ddot{\vec{r}} = -\mu \frac{\vec{r}}{r^3} \quad (1)$ where $\vec{r} \equiv \vec{x}_1 - \vec{x}_2$, $r = \vec{r}$ is the norm of $\vec{r}$ and $\mu = G(m_1 + m_2)$. Here, $\dot{G} \equiv \frac{dG(t)}{dt}$ for any time dependent variable G. Therefore $\frac{d^2 r}{dt^2} = \ddot{r}$.	1 pts
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Take the dot product of (1) with $\dot{\vec{r}}$ to show that:

C	$\frac{d}{dt} \underbrace{\left(\frac{(\dot{\vec{r}})^2}{2} - \frac{\mu}{r} \right)}_{\xi} = 0 \quad (2)$ ξ is an energy per unit mass; however, let us call it energy anyway.	1.5 pts
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D	Next consider the quantity called angular momentum, defined as $\vec{\Lambda} \equiv \vec{r} \times \dot{\vec{r}}$. Show that it is a constant for this system.	1.5 pts
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Consequently, the dynamics is now confined to a two-dimensional plane (perpendicular to the angular momentum vector), and we can adopt polar coordinates involving an orthogonal set of unit vectors ($\vec{e}_r, \vec{e}_\theta$) such that $\vec{r} = r\vec{e}_r$. One can project $\dot{\vec{r}}$ into this basis and show that the dynamics is simplified to a two-dimensional plane. Thus one can utilize polar coordinates.

E	Use this basis to show that $\xi = \frac{\Lambda^2}{2} \left(\left(\frac{dr}{d\theta} \right)^2 + r^2 \right) \frac{1}{r^4} - \frac{\mu}{r} \quad (3)$	2.5 pts
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Use the change of variable $u \equiv \frac{1}{r} - \frac{\mu}{\Lambda^2}$ to find the differential equation

F
$$\left(\frac{du}{d\theta}\right)^2 = \alpha^2 - u^2 \quad (4) \quad 2.5 \text{ pts}$$

where $\alpha^2 = \frac{2\xi}{\Lambda^2} + \frac{\mu^2}{\Lambda^4}$.

Solve (4) by integration between $u(\theta = 0) = 0$ and $u(\theta < \pi)$. One could use $\text{Arccos}'(x) = \frac{-1}{\sqrt{1-x^2}}$ for any $x \in [-1, 1]$. Do the same for $\theta \in [\pi, 2\pi]$. Also, show that this equation represents a conic equation in polar coordinates:

G
$$\frac{P}{r} = 1 + e \cos \theta \quad 3 \text{ pts}$$

where,

$$P = \frac{\Lambda^2}{\mu} \quad \& \quad e = \sqrt{1 + \frac{2\xi\Lambda^2}{\mu^2}}$$

Problem 2: Moving Electrons in Metallic Conductors [12 Points]

Assume that the electrons moving in a conductor act all *spooky*. Let us elaborate on what this *spooky* business is.

- First consider a single atom. The atom has a positive core and a negative electron around it. In our system, the positive core doesn't move, they remain static. Only the electrons flow around the static positive core. We can consider this flow of electrons as the flow of gas.
- The electrons are like billiard balls which are moving randomly in the matter. One may think that since electrons are negatively charged, they will repel each other. But spookily or miraculously enough, these balls don't interact electrically with each other. What's more, they can pass through each other without colliding! The only thing is that they can collide with are the stationary positive cores, and also the surface of the conductor (so that they don't fly away from the matter), and these collisions are instantaneous.
- The final velocity of an electron after a collision with the core is not related to the initial velocity before the collision. The final velocity is random.
- There is a characteristic time τ which has the property that, the probability that an electron will undergo another collision within a time Δt after a collision is given $\frac{\Delta t}{\tau}$ if $\Delta t \leq \tau$, and exactly 1 if $\Delta t \geq \tau$. Let us call this time the relaxation time.
- Under an external force $\mathbf{f}(t)$ or under the lack of it, the electrons are going to follow Newton's laws.

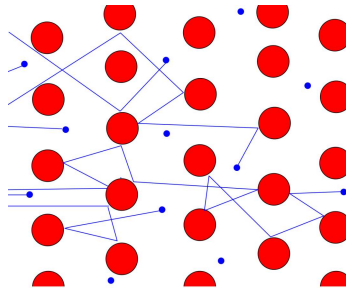


Figure 1: The thin lines represent the electron's trajectory and the solid balls are the positive cores.

You have the mass of an electron to be m , the charge of an electron to be $-e$, the mass density of the matter (weight of the element per volume) to be ρ_m , the atomic mass of the conducting element to be A , the Avogadro number to be N_A , the number of moving electrons available per atom to be Z , and the number of moving electrons per unit volume to be n .

Part A: Finding the Density

A1	Find an expression for n in terms of ρ_m , A , N_A and Z .	1 pts
A2	Show that by taking the dimensions of both sides of the equations you get a match.	1 pts

Part B: Average Momentum

There are two types of electrons between the time interval t to $t + \Delta t$: ones which go through one or more collisions with the cores, and the others which do not go through any collision at all.

Let us denote the average momentum per electron at time t by $\langle \mathbf{p}(t) \rangle_{\text{all}}$ and that at the time $t + \Delta t$ by $\langle \mathbf{p}(t + \Delta t) \rangle_{\text{all}}$. These averages are taken over all the electrons.

When we are only considering the electrons that have had a collision, we write the average momenta by $\langle \mathbf{p}(t) \rangle_{\text{col}}$ and $\langle \mathbf{p}(t + \Delta t) \rangle_{\text{col}}$, and when we are only considering the electrons that did not collide, we write $\langle \mathbf{p}(t) \rangle_{\text{noncol}}$ and $\langle \mathbf{p}(t + \Delta t) \rangle_{\text{noncol}}$.

We learnt before that τ is the characteristic time. And the probability of an electron colliding with the core within a time Δt is $\frac{\Delta t}{\tau}$. Assume that an external force $\mathbf{f}(t)$ is present.

B1	Find an expression that relates $\langle \mathbf{p}(t + \Delta t) \rangle_{\text{all}}$, $\langle \mathbf{p}(t + \Delta t) \rangle_{\text{col}}$, and $\langle \mathbf{p}(t + \Delta t) \rangle_{\text{noncol}}$. [Hint: You have to use the idea of a weighted average. See the appendix.]	1.5 pts
B2	Using Taylor expansion, express $\langle \mathbf{p}(t + \Delta t) \rangle_{\text{col}}$, and $\langle \mathbf{p}(t + \Delta t) \rangle_{\text{noncol}}$ in terms of $\mathbf{f}(t)$, Δt , $\langle \mathbf{p}(t) \rangle_{\text{col}}$, and $\langle \mathbf{p}(t) \rangle_{\text{noncol}}$, and put these expressions for $\langle \mathbf{p}(t + \Delta t) \rangle_{\text{col}}$, and $\langle \mathbf{p}(t + \Delta t) \rangle_{\text{noncol}}$ in the expression that you got in part B1. [Hint: See the appendix.]	1.5 pts
B3	Now, assume that Δt is very small so that we can ignore the higher order terms in Δt , i.e. terms containing $(\Delta t)^2$, $(\Delta t)^3$, and so on. Only the terms not containing Δt , or containing only the first power of it will survive this approximation. After this approximation, and using $\langle \mathbf{p}(t) \rangle_{\text{all}} = \langle \mathbf{p}(t) \rangle_{\text{noncol}}$, the expression in part B3 can be written as a differential equation in the limit $\Delta t \rightarrow 0$. Find this differential equation.	2 pts

Part C: Solving the Differential Equation

When the net force is zero, we call it a steady state. Consider the steady-state solution of a differential equation. Now we have the differential equation that looks like the following

$$\frac{d}{dx} A(x) = kA(x)$$

C1	Which one of the following do you think would be a solution to this equation? (a) $A_1(x) = C_1 \sin(kx)$ (b) $A_2(x) = k \ln(x) + C_2$ (c) $A_3(x) = C_3 e^{kx}$ [Hint: $\frac{d}{dx} A(x)$ measures the slope of the tangent line of $A(x)$ at x, and consult your graphing skills.]	1 pts
C2	Then what will be the solution for the equation derived in B3.	1 pts
C3	Draw a qualitative graph of p vs t .	2 pts

Part D: Playing with dimensions

The dimension of electrical resistance R is

$$[R] = \frac{ML^2}{T^3 I^2}$$

We know the resistivity ρ is the electrical resistance of a conductor of unit cross-sectional area and unit length. Assume that we can write the characteristic time τ as a function of the mass of an electron m , the resistivity of the conductor ρ , the number of electrons per unit volume n , and the magnitude of the charge of an electron e .

D1	Now derive the formula for τ by dimensional analysis.	1 pts
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Appendix

Weighted Average

A weighted average is the average of values which are scaled by importance. The weighted average equals the sum of weights times values divided by the sum of the weights.

If x_1, x_2, x_3 are the values and w_1, w_2, w_3 are the weights, then the weighted average will be

$$\text{Weighted Average} = \frac{w_1 x_1 + w_2 x_2 + w_3 x_3}{w_1 + w_2 + w_3}$$

Taylor Series

If f is a function of x , then the *Taylor expansion* around of $f(x_0)$ is

$$f(x)|_{x=x_0} = f(x_0) + \frac{df(x_0)}{dx} \cdot \frac{x}{1!} + \frac{d^2 f(x_0)}{dx^2} \cdot \frac{(x)^2}{2!} + \dots + \frac{d^n f(x_0)}{dx^n} \cdot \frac{(x)^n}{n!} + \dots$$